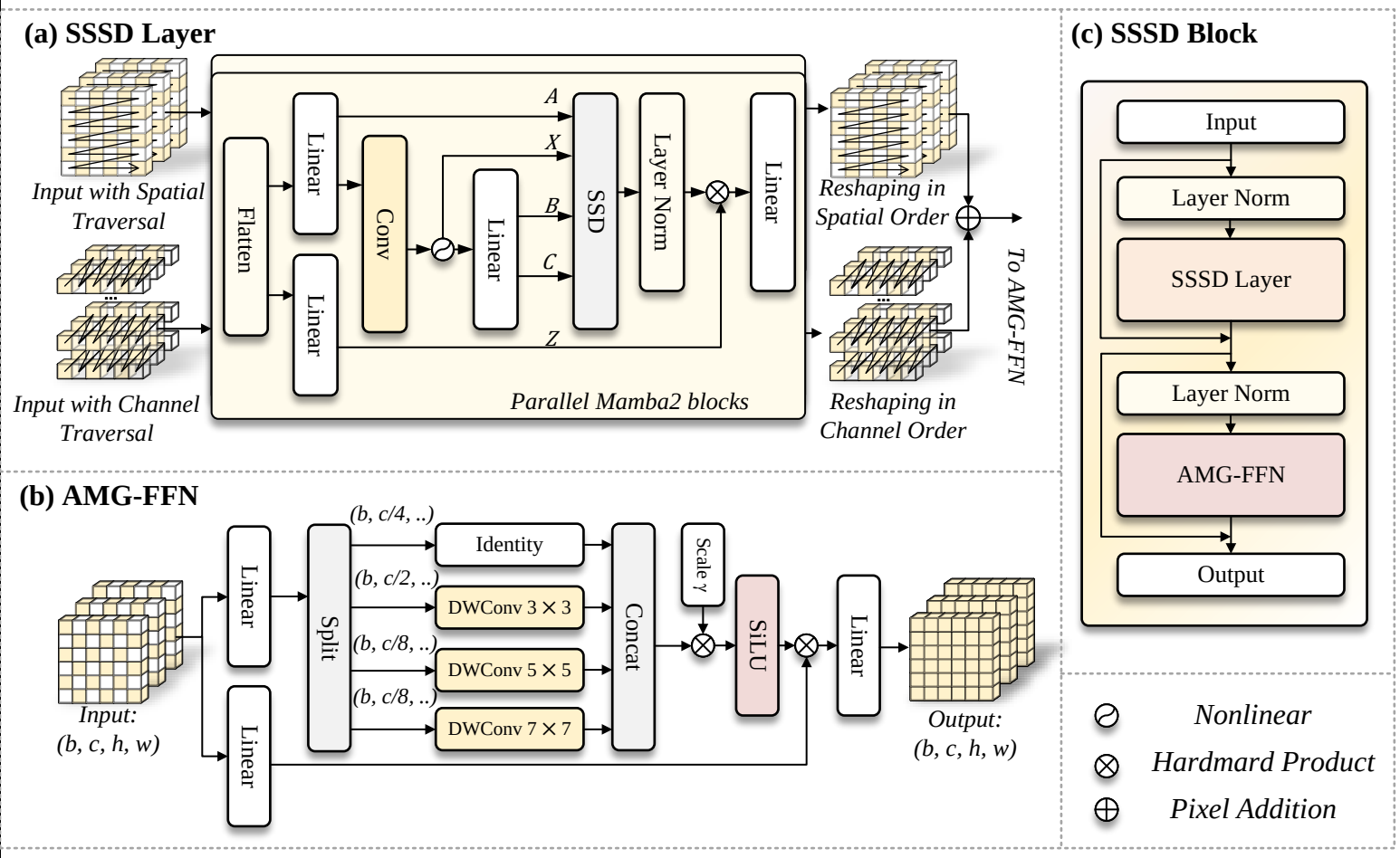
(a) SSSD Layer
Input with Spatial Traversal
Input with Channel Traversal
Flatten
Linear
Linear
Conv
Linear
A
X
B
C
Z
SSD
Layer Norm
Linear
Parallel Mamba2 blocks
Reshaping in Spatial Order
Reshaping in Channel Order
To AMG-FFN
(b) AMG-FFN
Input: (b, c, h, w)
Linear
Linear
Split
(b, c/4, ..)
(b, c/2, ..)
(b, c/8, ..)
(b, c/8, ..)
Identity
DWConv 3 × 3
DWConv 5 × 5
DWConv 7 × 7
Concat
Scale γ
SiLU
Linear
Output: (b, c, h, w)
(c) SSSD Block
Input
Layer Norm
SSSD Layer
Layer Norm
AMG-FFN
Output
Nonlinear
Hardmard Product
Pixel Addition